A Coherent-Staple Good-Collar Theorem

Uniform small-time conditional curvature after the frustrated-boundary obstruction

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Abstract

Worst-case conditioning of a Yang–Mills link on arbitrary surrounding links does not preserve small plaquette curvature: five identity staples and one negative-center staple move the conditional weak-coupling minimum away from the identity. We prove a boundary-restricted replacement. If the m staples lie in a common geodesic ball of radius η in a compact group with bi-invariant metric, then the conditional heat-kernel measure has, for every target plaquette outside curvature radius ρ, a small-time rate gap at least $\rho^2 - m\eta^2$. Thus $\eta < \rho/\sqrt{m}$ is a uniform sufficient condition. The condition is gauge covariant and excludes the center-frustrated boundary, whose minimum common-ball radius is $\pi/2$ in SU(2). The result supplies the good-collar half of a two-tier activity argument; the activity of collar failure remains a separate obligation.

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1 Conditional geometry

Let $G = \mathrm{SU}(2)$ with bi-invariant geodesic distance d and identity I. Orient the m plaquettes incident on a selected link so that their conditional factors are $p_t(US_i)$, where S_i is the corresponding staple. The conditional measure is

$$\gamma_{t,S}(\mathrm{d}U) = Z_{t,S}^{-1} \prod_{i=1}^{m} p_t(US_i)\,\mathrm{d}U. \tag{1}$$

For SU(2) with heat generator $\frac{1}{2}\Delta$, the exponential scale is

$$2t \log p_t(g) \longrightarrow -d(I,g)^2. \tag{2}$$

Proposition 1.1 (Gauge-covariant staple coherence)**.** *Suppose there is $C \in G$ such that*

$$\max_{1\le i\le m} d(C, S_i) \le \eta. \tag{3}$$

The least such η is invariant under the endpoint gauge action on all staples.

Proof. The endpoint action is a common left and right multiplication of the staple family. A bi-invariant metric preserves all distances under this action and transports a common center C to the transformed common center. □

Set $V = UC$ and $Q_i = C^{-1}S_i$. Then $d(I, Q_i) \le \eta$ and the conditional rate energy becomes

$$E_Q(V) = \sum_{i=1}^{m} d(I, VQ_i)^2. \tag{4}$$

2 Uniform rate theorem

For target j, write

$$B_j(\rho) = \{V : d(I, VQ_j) \ge \rho\}.$$

Theorem 2.1 (Coherent-staple conditional rate)**.** *For every coherent staple family satisfying* (3)*,*

$$\inf_{B_j(\rho)} E_Q - \min_G E_Q \ge \rho^2 - m\eta^2. \tag{5}$$

Consequently, if $\eta < \rho/\sqrt{m}$, then

$$\limsup_{t\downarrow 0} 2t \log \gamma_{t,S}(B_j(\rho)) \le -(\rho^2 - m\eta^2) < 0. \tag{6}$$

The same rate upper bound holds for the union of the finitely many target events.

Proof. At the reference point $V = I$,

$$E_Q(I) = \sum_i d(I, Q_i)^2 \le m\eta^2,$$

so $\min E_Q \le m\eta^2$. On $B_j(\rho)$, the jth nonnegative summand in (4) is at least ρ^2, hence $\inf_{B_j(\rho)} E_Q \ge \rho^2$. Subtraction proves (5). The heat-kernel Laplace principle from (2) gives (6). A finite union does not change the exponential rate upper bound. □

Remark 2.2. *The theorem uses a deliberately robust energy comparison. It does not claim that $\rho/\sqrt{m}$ is the optimal coherence threshold. Its value is that it is uniform over all noncommuting staple families and needs no coordinate choice.*

3 The center-frustrated boundary is excluded

The geodesic distance between I and $-I$ in SU(2) is π. Any ball containing both has radius at least $\pi/2$, and the midpoint ball attains this value. Thus the Stage 10 family containing both center elements has coherence radius exactly $\pi/2$. It cannot enter a collar with ultraviolet $\eta \ll 1$.

For comparison, put all coherent staples in a common one-parameter subgroup and write their signed angles as $q_i \in [-\eta, \eta]$. In the convex chart,

$$E(\theta) = \sum_i (\theta + q_i)^2, \qquad \theta_* = -\frac{1}{m}\sum_i q_i.$$

Enumeration of the 2^m vertices of the angle box gives the sharp commuting bound

$$\max_i |\theta_* + q_i| \le \frac{2(m-1)}{m}\eta. \tag{7}$$

For $m = 6$ this is $5\eta/3$, smaller than the general metric certificate $\sqrt{6}\eta$.

4 Ultraviolet calibration

At the frozen Stage 09 parameters and the declared collar radius:

quantity	value
t	3×10^{-6}
m	6
ρ	0.0082
η	0.001
$\rho/\sqrt{m}$	0.00334763598
$\rho^2 - m\eta^2$	0.00006124
$(\rho^2 - m\eta^2)/(2t)$	10.2066667
exponential proxy	3.69233×10^{-5}

The proxy is a calibration of the asymptotic rate, not a finite-t prefactor theorem.

5 Frontier audit

The result proves that good-collar conditioning is mathematically viable and that the exact center obstruction is outside its hypothesis. It does not estimate how often the collar fails under the interacting link measure. A full-measure argument must add a connected activity for incoherent collars, including center, boundary-holonomy, orbit-atlas, and topological labels. No repeated RG, continuum construction, or Hamiltonian mass gap follows from (6). The result is a restricted theorem at claim level C2.